

Mechanics of Solids (3130608)

Unit – 2

Application of fundamentals of statics



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Syllabus – Module 2 Description



GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering

Subject Code: 3130608

Semester III

Mechanics of Solids

MODULE 2

Applications of fundamentals of statics

Statically determinate beams:

Types of loads, Types of supports, Types of beams;
Determination of support reactions, Relationship between loading, shear force & bending moment, Bending moment and shear force diagrams for beams subjected to only three types of loads :i) concentrated loads ii) uniformly distributed loads iii) couples and their combinations; Point of contraflexure, point & magnitude of maximum bending moment, maximum shear force

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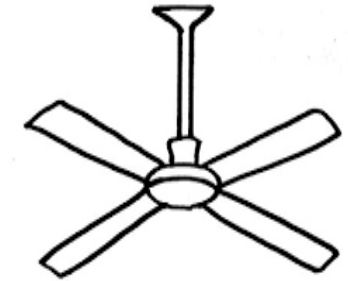
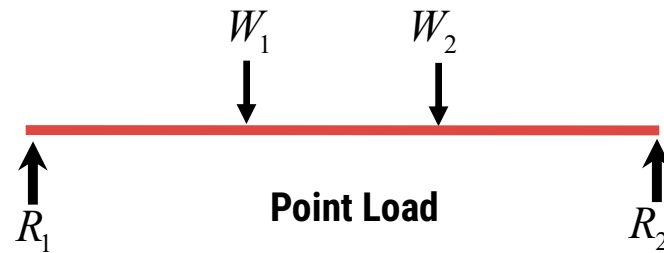
Topics covered in Presentation

- ▶ Types of Load
- ▶ Types of Support
- ▶ Types of Beam
- ▶ Sign Conventions for Forces & Moment
- ▶ Equilibrium Condition
- ▶ Numerical

Types of Load

Load concentrated on a very small length compare to the length of the beam, is known as point load or concentrated load. Point load may have any direction

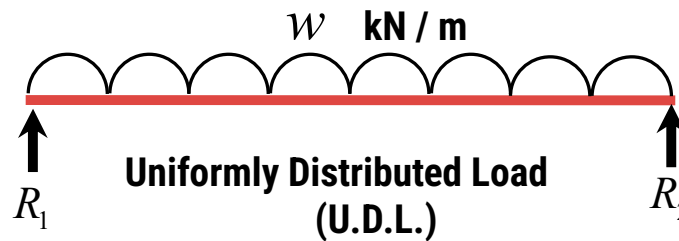
Truck transferring entire load of truck at 4 point of contact to the bridge is point load.



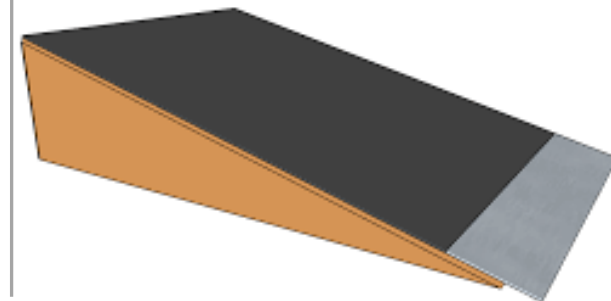
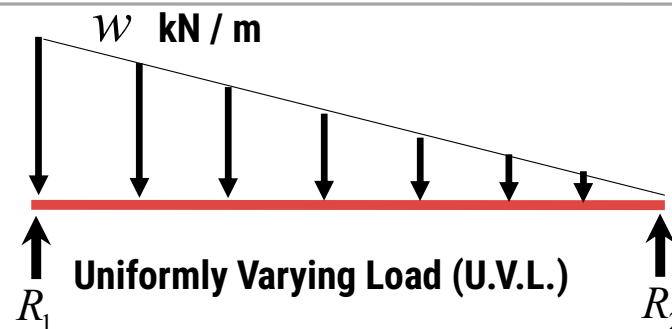
Load spread uniformly over the length of the beam is known as uniformly distributed load.

Water tank resting on the beam length

Pipe full of water in which weight of the load per unit length is constant

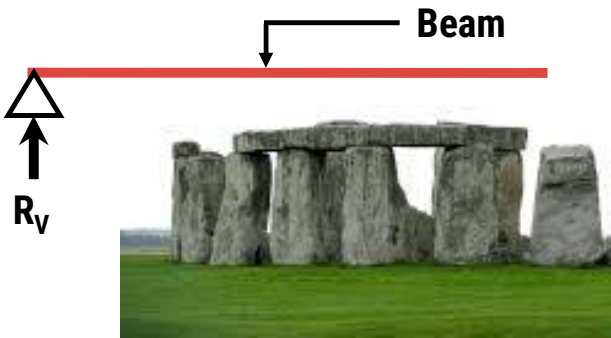


Load in which value of the load spread over the length if uniformly increasing or decreasing from one end to the other is known as uniformly varying load. It is also called triangular load



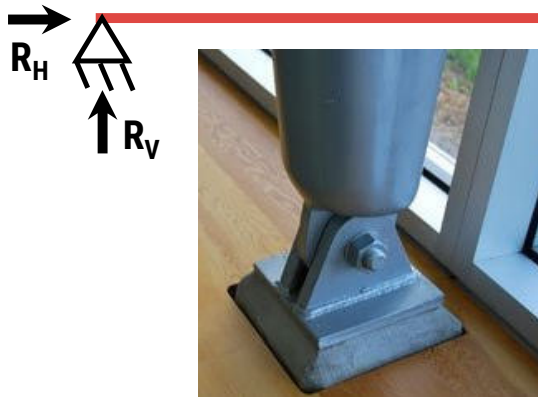
Type of Support

▶ SIMPLE SUPPORT



In this type of support beam is simply supported on the support. There is no connection between beam and support. Only vertical reaction will be produced

▶ HINGE SUPPORT



Beam and support are connected by a hinge. Beam can rotate about the hinge. Reaction may be vertical, horizontal or inclined.

R_H = Horizontal Reaction

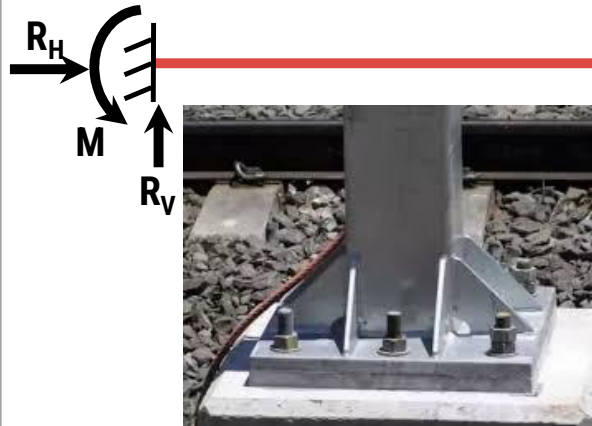
▶ ROLLER SUPPORT



Here rollers are placed below beam and beam can slide over the rollers. Reaction will be perpendicular to the surface on which rollers are supported.

This type of support is normally provided at the end of a bridge

▶ FIXED SUPPORT

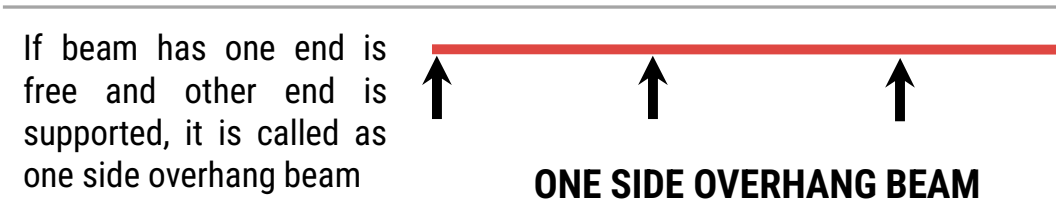
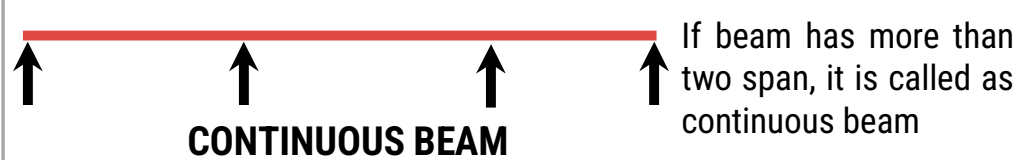
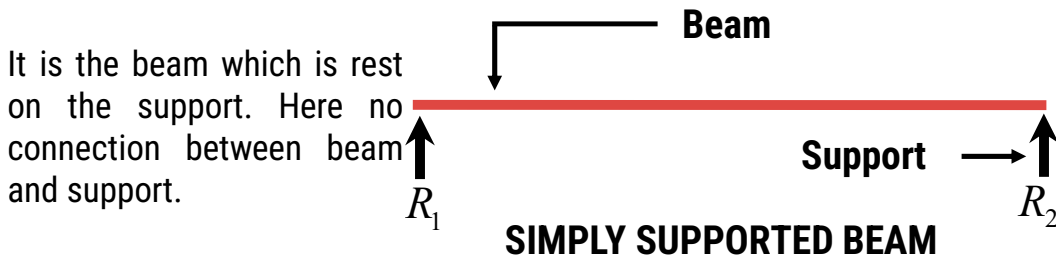


Beam is completely fixed at end in the wall or support. Beam cannot rotate at end. Reactions may be vertical, horizontal, inclined & moment

R_v = Vertical Reaction

M = Moment

Types of Beam



If one end of beam is fixed and other is supported with prop then it is known as propped cantilever beam.

Sign Conventions For Forces & Moment

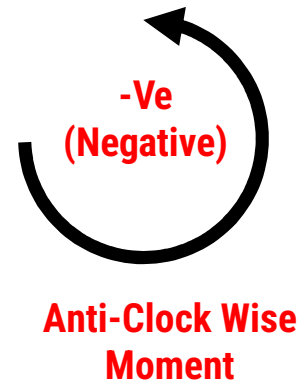
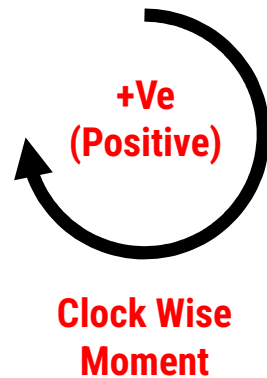
► Vertical Force



► Horizontal Force



► Moment



Equilibrium Condition

- ▶ Summation of Horizontal force is zero

$$\sum H = 0$$

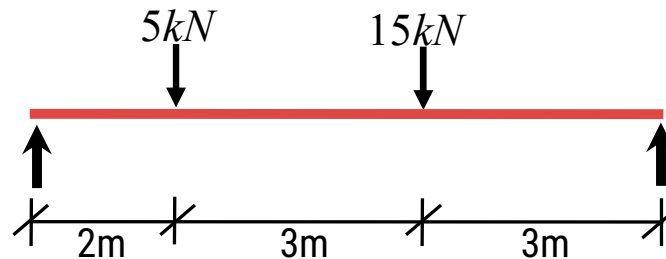
- ▶ Summation of Vertical force is zero

$$\sum V = 0$$

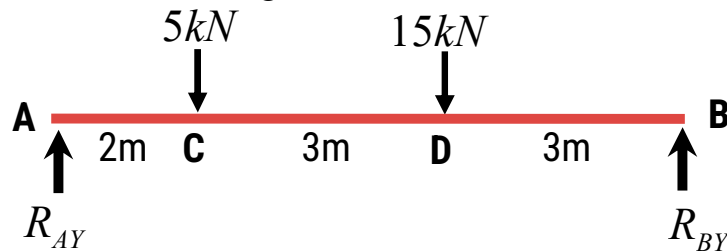
- ▶ Summation of All Moment at any particular point is zero

$$\sum M_A = 0$$

Ex - 1 Find Support Reaction of Beam as Shown in Figure



Answer : - From a given data



► Applying equilibrium condition

$$\Sigma V = 0 \quad \uparrow +\text{Ve (Positive)}$$

$$R_{AY} + R_{BY} - 5 - 15 = 0$$

$$R_{AY} + R_{BY} = 20 \dots \dots \dots \text{eqn(1)}$$

► Applying 2nd equilibrium condition

► Taking moment at point "A"

$$\Sigma M_A = 0 \quad \curvearrowright +\text{Ve}$$

$$\curvearrowleft + 5 \times 2 \quad \curvearrowleft + 15 \times (2 + 3) \quad \curvearrowright - R_{BY} \times (2 + 3 + 3) = 0$$

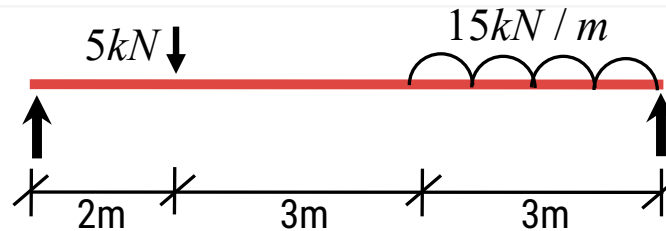
$$+ 5 \times 2 + 15 \times 5 - R_{BY} \times 8 = 0$$

$$\therefore \frac{5 \times 2 + 15 \times 5}{8} = R_{BY} \Rightarrow R_{BY} = 10.625 \text{ kN} \quad \uparrow$$

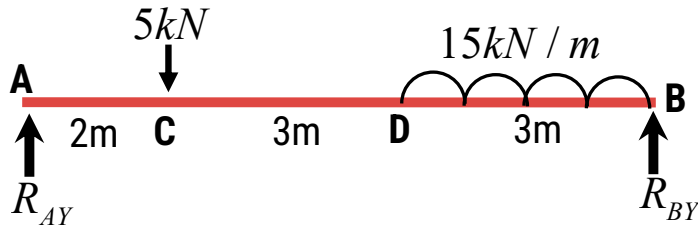
► From equation (1)

$$R_{AY} = 9.375 \text{ kN} \quad \uparrow$$

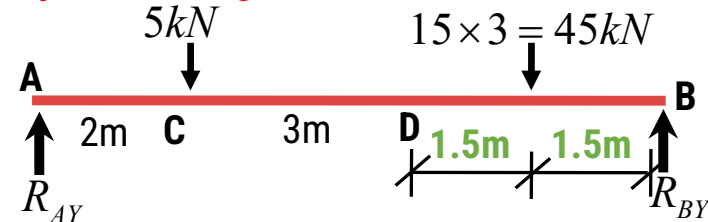
Ex - 2 Find Support Reaction of Beam as Shown in Figure



Answer : - From a given data



Equivalent Diagram



► Applying equilibrium condition

$$\sum V = 0 \quad \uparrow +\text{Ve (Positive)}$$

$$R_{AY} + R_{BY} - 5 - 15 \times 3 = 0$$

$$R_{AY} + R_{BY} = 50 \dots \dots \dots \text{eqn(1)}$$

► Applying 2nd equilibrium condition

► Taking moment at point "A"

$$\sum M_A = 0 \quad +\text{Ve}$$

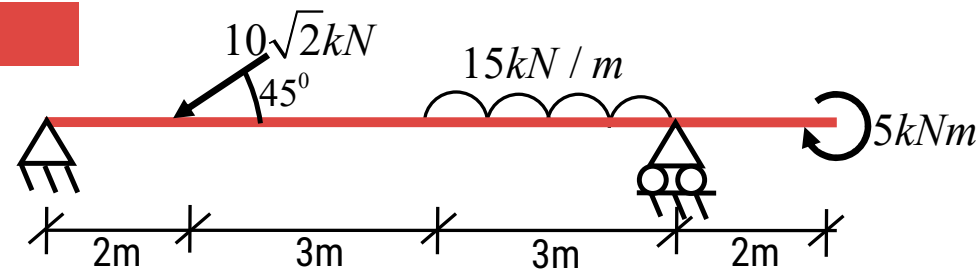
$$+5 \times 2 + 45 \times (2 + 3 + 1.5) - R_{BY} \times (2 + 3 + 3) = 0$$

$$+5 \times 2 + 45 \times 6.5 - R_{BY} \times 8 = 0$$

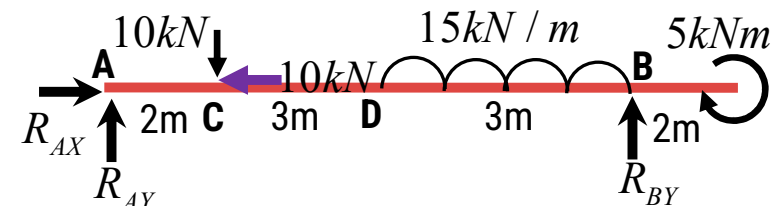
$$\therefore \frac{5 \times 2 + 45 \times 6.5}{8} = R_{BY} \Rightarrow R_{BY} = 37.81 \text{ kN} \quad \uparrow$$

► From equation (1) $R_{AY} = 12.19 \text{ kN} \quad \uparrow$

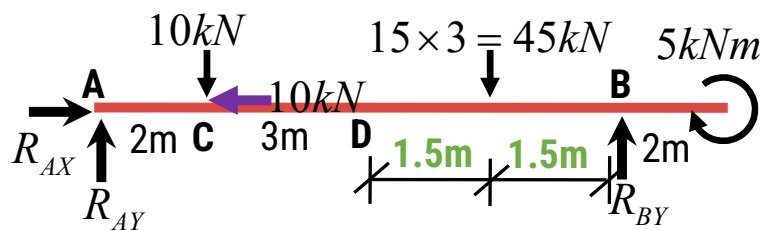
Ex - 3 Find Support Reaction of Beam as Shown in Figure



Answer : - From a given data



Equivalent diagram



► Applying 3rd equilibrium condition

► Taking moment at point "A"

► Applying equilibrium condition

$$\sum V = 0 \quad \uparrow +\text{Ve (Positive)}$$

$$R_{AY} + R_{BY} - 10 - 15 \times 3 = 0$$

$$R_{AY} + R_{BY} = 55 \dots \dots \dots \text{eqn(1)}$$

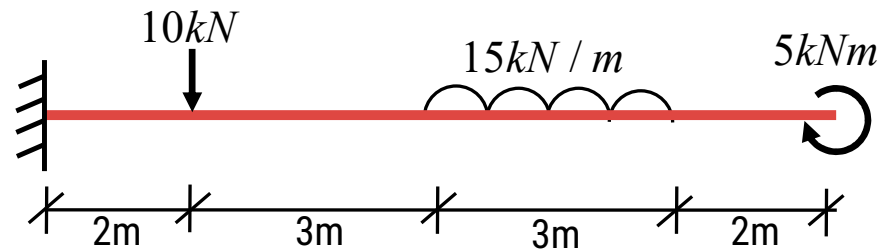
$$\sum M_A = 0 \quad \curvearrowright +\text{Ve}$$

$$\curvearrowleft + 10 \times 2 \quad \curvearrowleft + 45 \times (2 + 3 + 1.5) \quad \curvearrowleft + 5 \quad \curvearrowright + R_{BY} \times (2 + 3 + 3) = 0 \quad \Rightarrow \quad +5 \times 2 + 45 \times 6.5 - R_{BY} \times 8 = 0$$

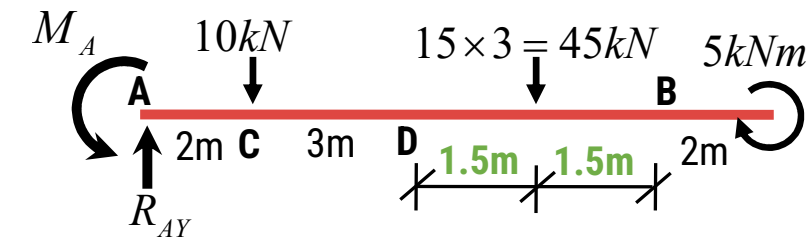
$$\therefore \frac{5 \times 2 + 45 \times 6.5 + 5}{8} = R_{BY} \Rightarrow R_{BY} = 39.69 \text{ kN} \quad \uparrow$$

$$R_{AY} = 15.31 \text{ kN} \quad \uparrow$$

Ex - 4 Find Support Reaction of Beam as Shown in Figure

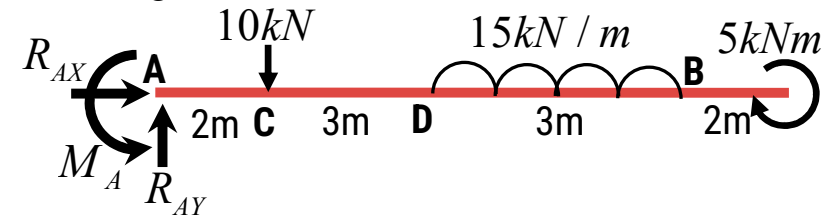


Equivalent diagram



Answer : -

From a given data



► Applying equilibrium condition

$$\sum V = 0$$

↑ +Ve (Positive)

$$R_{AY} - 10 - 15 \times 3 = 0 \Rightarrow R_{AY} = 55kN \uparrow$$

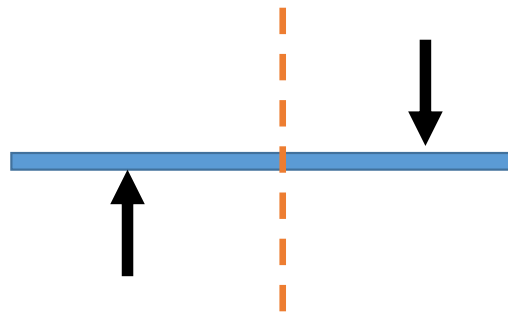
► Taking moment at point "A" $\sum M_A = 0$ +Ve

$$-M_A + 10 \times 2 + 45 \times (2 + 3 + 1.5) + 5 = 0 \Rightarrow 5 + 10 \times 2 + 45 \times 6.5 - M_A = 0$$

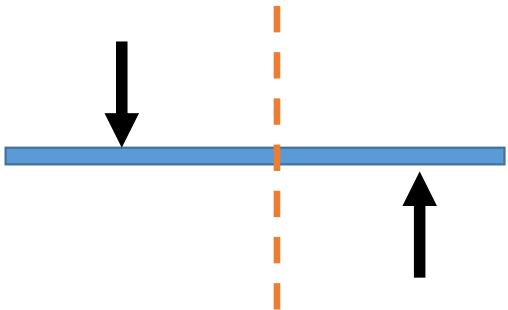
$$M_A = 317.5kN \cdot m$$

Sign conventional for shear force & bending moment diagram

Shear Force Diagram

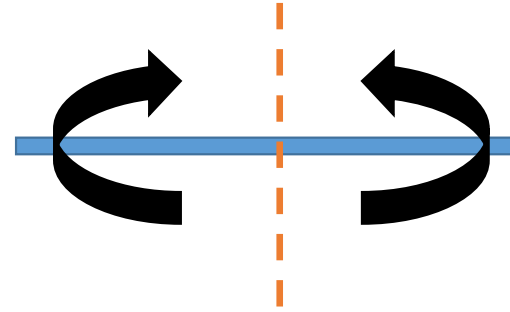


+ Ve (Positive)

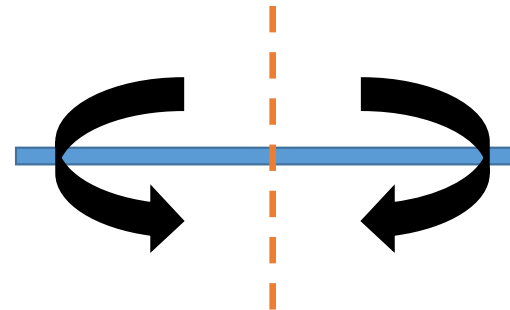


- Ve (Negative)

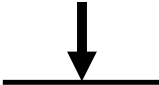
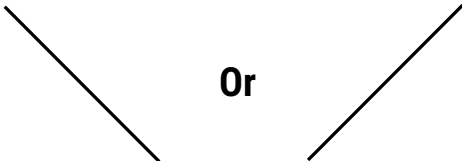
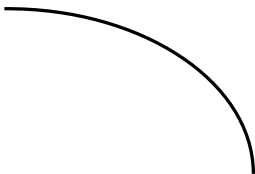
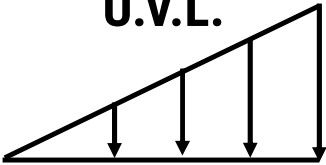
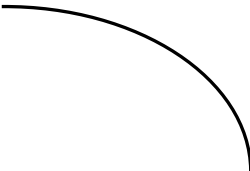
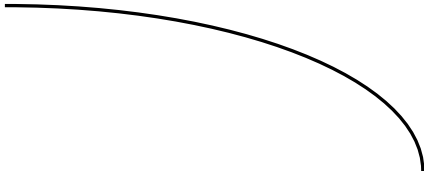
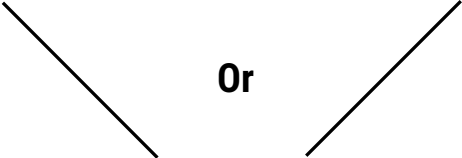
Bending Moment Diagram

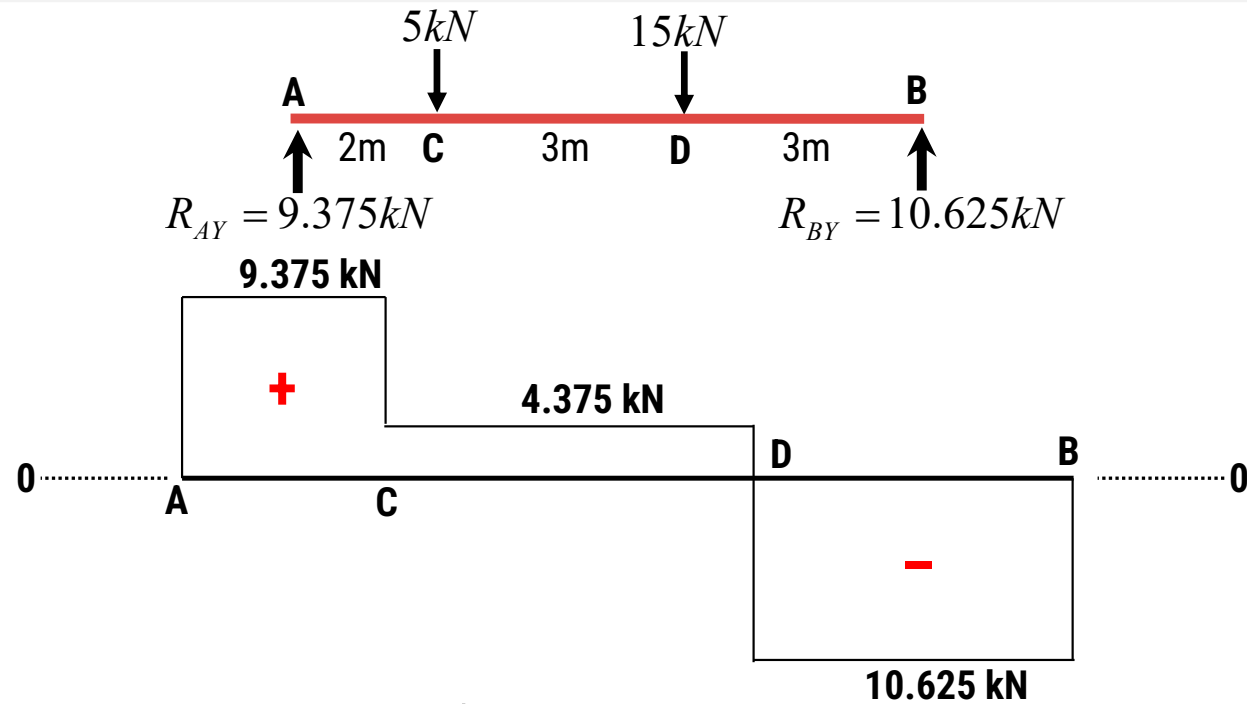


+Ve (Positive)



- Ve (Negative)

Forces	Shear force diagram	Bending moment diagram
Point Load 		Follow Previous
U.D.L. 		
U.V.L. 		
Moment 	Follow Previous	
No Load Condition		

Ex - 5**Draw shear force diagram and bending moment diagram of example no. 1**

$$S.F.@ A(L.H.S.) = 0\text{ kN}$$

$$S.F.@ C(L.H.S.) = +9.375\text{ kN}$$

$$S.F.@ D(L.H.S.) = 4.375\text{ kN}$$

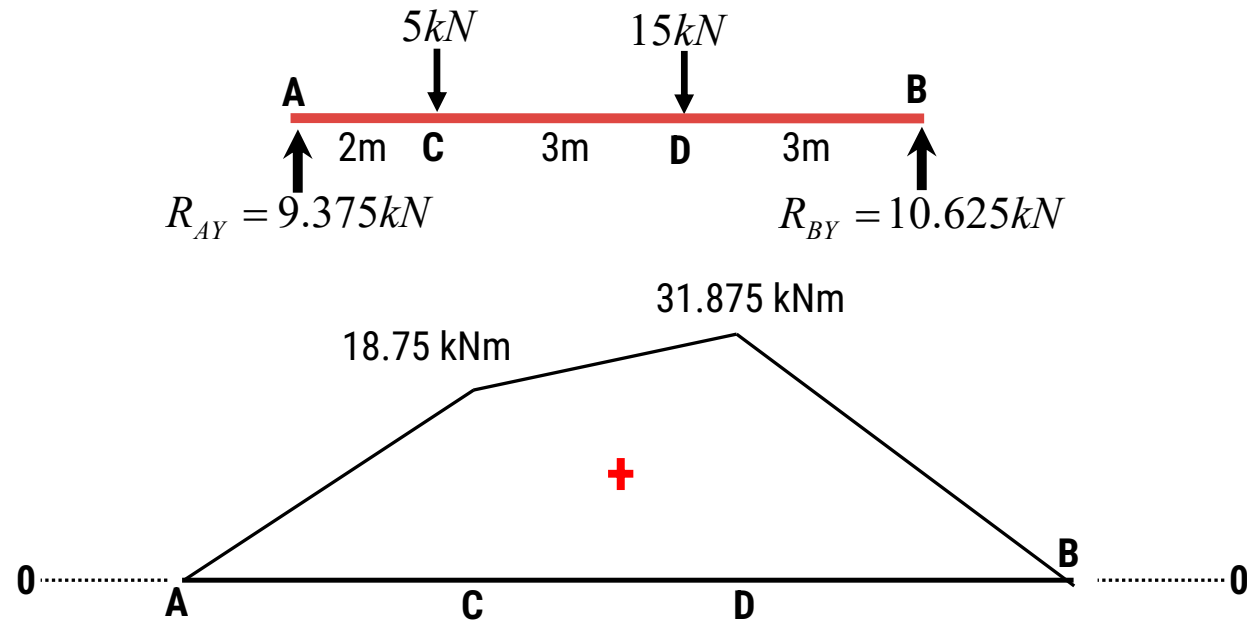
$$S.F.@ B(L.H.S.) = -10.625\text{ kN}$$

$$S.F.@ A(R.H.S.) = +9.375\text{ kN}$$

$$S.F.@ C(R.H.S.) = +9.375 - 5 = 4.375\text{ kN}$$

$$S.F.@ D(R.H.S.) = +4.375 - 15 = -10.625\text{ kN}$$

$$S.F.@ B(R.H.S.) = -10.625 + 10.625 = 0\text{ kN}$$



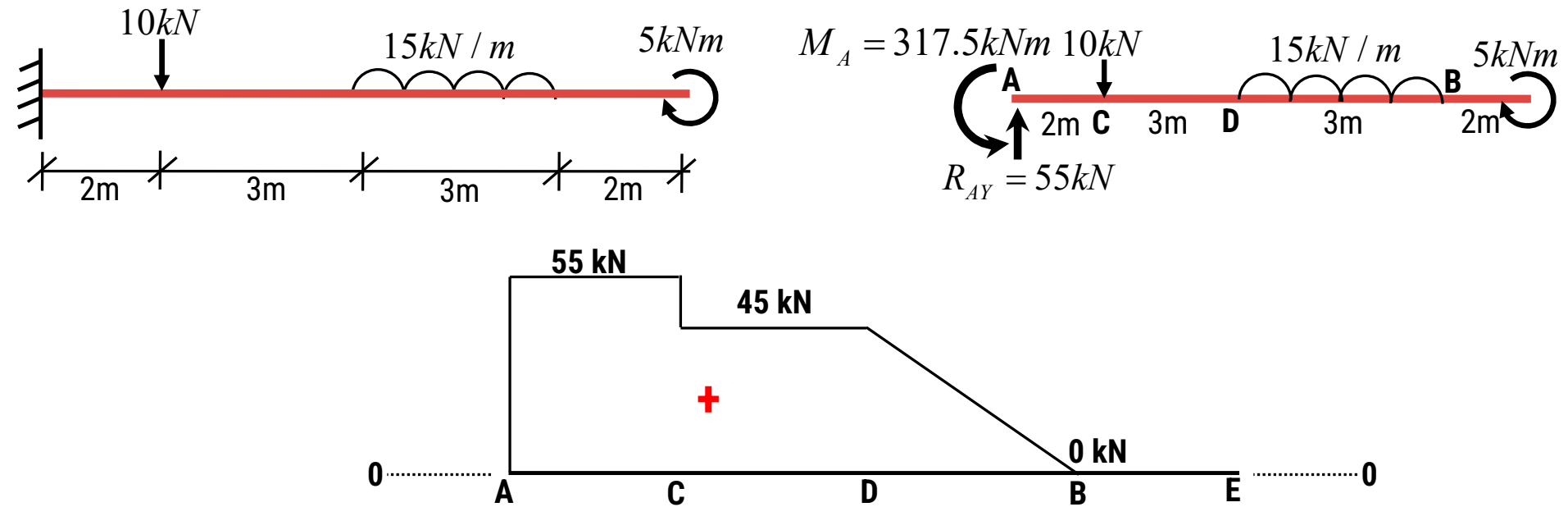
$$B.M.@ A = 9.375 \times 0 = 0\text{kNm}$$

$$B.M.@ B = 10.625 \times 0 = 0\text{kNm}$$

$$B.M.@ C = 9.375 \times 2 = 18.75\text{kNm}$$

$$B.M.@ D = 10.625 \times 3 = 31.875\text{kNm}$$

Ex - 6 Draw shear force diagram and bending moment diagram of example no. 4



$$S.F @ A(L.H.S.) = 0kN$$

$$S.F @ C(L.H.S.) = +55kN$$

$$S.F @ D(L.H.S.) = 45kN$$

$$S.F @ B(L.H.S.) = 45 - 15 \times 3 = 0kN$$

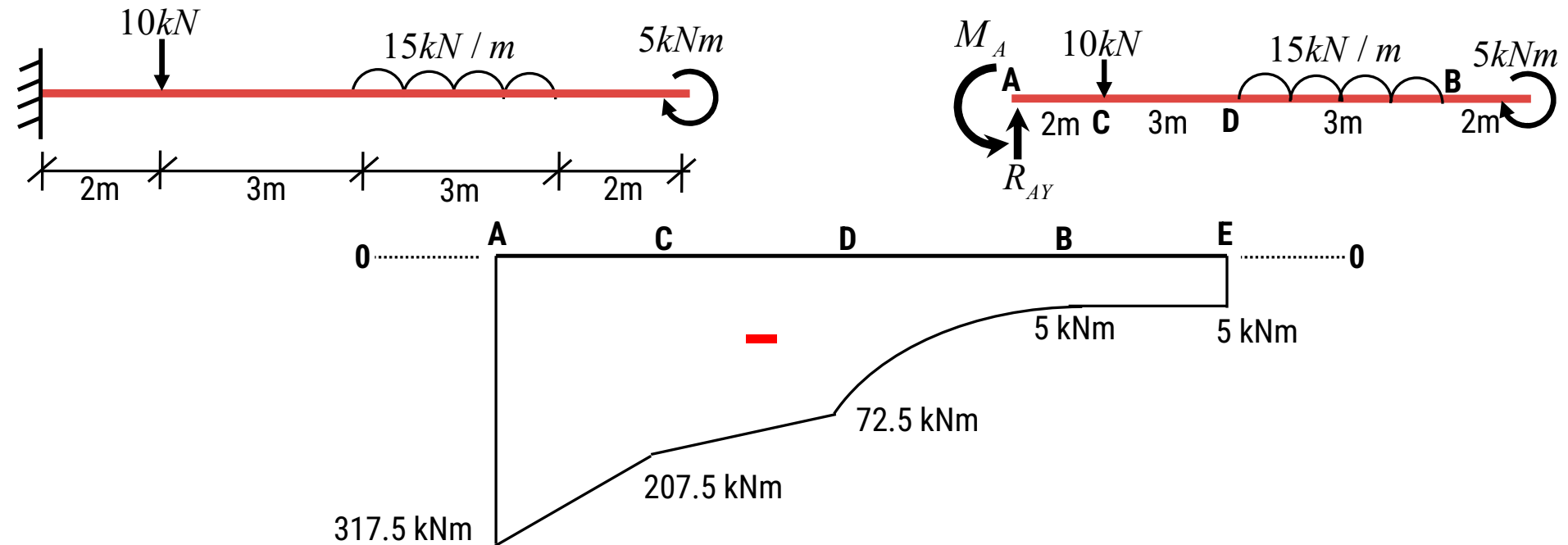
$$S.F @ A(R.H.S.) = +55kN$$

$$S.F @ C(R.H.S.) = 55 - 10 = 45kN$$

$$S.F @ D(R.H.S.) = 45kN$$

$$S.F @ B(R.H.S.) = 0kN$$

Ex - 6 Draw shear force diagram and bending moment diagram of example no. 4



$$B.M @ E(R.H.S.) = 0kNm$$

$$B.M @ E(L.H.S.) = -5kNm$$

$$B.M @ B = -5kNm$$

$$B.M @ D = -5 - 15 \times 3 \times 1.5 = -72.5kNm$$

$$B.M @ C = -5 - 15 \times 3 \times (3 + 1.5) = -207.5kNm$$

$$B.M @ A(R.H.S.) = -5 - 15 \times 3 \times (3 + 1.5 + 2) - 10 \times 2 = -317.5kNm$$

$$B.M @ A(L.H.S.) = -317.5 + 317.5 = 0kNm$$